

Lecture-12

Relative Maximum: A function $f: S \subset \mathbb{R} \rightarrow \mathbb{R}$ is said to have a relative maximum at $c \in S$ if $\exists$ a open interval I containing c such that

$$f(x) \leq f(c) \quad \forall x \in I \cap S$$

Similarly, for local/relative minimum,

$$f(x) \geq f(c) \quad \forall x \in I \cap S$$

$\rightarrow$ A number at which a relative maximum/minimum is obtained is called a local extreme value or a local extremum of f .

Thm: Let f be defined on an open interval I , and assume that f has a relative maximum/minimum at an interior point $c \in I$.

$\rightarrow$ If the function f is continuous on I and is differentiable on I , then $f'(c) = 0$.

Proof: Sign preserving property and defn. of derivative.

Rolle's Theorem Let $a < b$, $a, b \in \mathbb{R}$

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function that is differentiable on (a, b) . Also assume that $f(a) = f(b)$. Then $\exists$ at least one point $c \in (a, b)$ such that

$$f'(c) = 0$$

Mean Value Theorem (MVT) Let $a < b$, $a, b \in \mathbb{R}$

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function that is differentiable on (a, b) . Then $\exists$ at least one $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

MVT $\Rightarrow \exists$ a point $c \in (a, b)$ such that the slope of tangent to the graph of f at c = slope of chord joining line points $(a, f(a))$ and $(b, f(b))$

Proof of MVT using Rolle's Theorem

Define $h: [a, b] \rightarrow \mathbb{R}$ by $h(x) = f(x)(b-a) - x(f(b) - f(a))$

- h is continuous on $[a, b]$
- h is differentiable on (a, b)
- $h(a) = h(b) = b f(a) - a f(b)$

By Rolle's Theorem applied to h , we get $\exists c \in (a, b)$ such that $h'(c) = 0$.

$$h'(c) = f'(c)(b-a) - (f(b) - f(a)) = 0$$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Warning $f(x) = |x|$, $x \in [-1, 1]$

$f(-1) = f(1)$ but $\nexists c \in (-1, 1)$ such that $f'(c) = 0$.

But f is not diff. on one point in $(-1, 1)$. So MVT/Rolle's Theorem will fail.

Thm Let $a < b$ and $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function such that f is diff. on (a, b) . Then we have:

- If $f'(x) > 0 \quad \forall x \in (a, b)$, then f is strictly increasing on $[a, b]$.
- If $f'(x) < 0 \quad \forall x \in (a, b)$, then f is strictly decreasing on $[a, b]$.
- If $f'(x) = 0 \quad \forall x \in (a, b)$, then f is a constant function on $[a, b]$.

Proof (i) To show $f(x) < f(y) \quad \forall a \leq x < y \leq b$
Applying Mean Value Theorem on f restricted to $[x, y]$ we get $\exists c \in (x, y)$ such that $f(y) - f(x) = f'(c)(y - x)$
 $\Rightarrow f(y) - f(x) > 0$
 $f(y) > f(x)$

Thm: Let $a, b \in \mathbb{R}$ such that $a < b$. Let $f: [a, b] \rightarrow \mathbb{R}$ be a cont. function such that f is diff. ble on (a, b) , except possibly at a point $c \in (a, b)$.

- i. If $f'(x) > 0 \ \forall x \in (a, c)$ and $f'(x) < 0 \ \forall x \in (c, b)$ then f has a local max. at c .
- ii. If $f'(x) < 0 \ \forall x \in (a, c)$ and $f'(x) > 0 \ \forall x \in (c, b)$ then f has a local min. at c .

Thm: Let $f: [a, b] \rightarrow \mathbb{R}$ be a cont. function such that f is twice diff. on (a, b) . Let c be a critical point of f so that $f'(c) = 0$. Then we have:

- i. If $f''(x) < 0 \ \forall x \in (a, b)$, then f has a local maximum at c .
- ii. If $f''(x) > 0 \ \forall x \in (a, b)$, then f has a local minimum at c .

a. Suppose $a < b$ and $f: [a, b] \rightarrow \mathbb{R}$ is a constant fn. such that f is diff. ble on (a, b) . Suppose $\exists c \in \mathbb{R}$ s.t. $f'(x) = c \ \forall x \in (a, b)$. Using MVT, show that $\exists d \in \mathbb{R}$ such that $f(x) = cx + d \ \forall x \in [a, b]$.

b. Suppose $a < b$ and $g: [a, b] \rightarrow [0, \infty)$ is a cont. fn. s.t. g is diff. ble on (a, b) . Suppose $\exists c \in \mathbb{R}$ s.t. $g'(x) = cg(x) \ \forall x \in (a, b)$. Using (i), show that $\exists A \in (0, \infty)$ s.t. $g(x) = Ae^{cx} \ \forall x \in [a, b]$.

Thm: Let $f: [a, b] \rightarrow \mathbb{R}$ be a cont. fn. s.t. f is twice continuously diff. on (a, b) [i.e., f'' is cont. on (a, b)]. Let $c \in (a, b)$ be a critical point, i.e., $f'(c) = 0$. Then we have

- If $f''(c) < 0$, then f has a local maxima at c .
- If $f''(c) > 0$, then f has a local minima at c .

Remark $f''(c) = 0$, then nothing can be said about local maxima/minima at c .

Taylor's Theorem Let $k \in \mathbb{N}$, and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be k -times diff.ble at $a \in \mathbb{R}$. Then $\exists$ a f_n $R_k: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(k)}(a)}{k!}(x-a)^k + R_k(x)$$

where $\lim_{x \rightarrow a} \frac{R_k(x)}{(x-a)^k} = 0$

Moreover, if $x \mapsto f$ is $(k+1)$ times diff.ble on (a, x) and $f^{(k)}(x)$ is continuous on the closed interval $[a, x]$, then

$$R_k(x) = \frac{f^{(k+1)}(\eta)}{(k+1)!}(x-a)^{k+1}$$

for some $\eta \in (a, x)$

$\rightarrow$ $f(x)$ without $R_k(x)$ is called k^{th} Taylor polynomial for $f(x)$ around a .

8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice continuously diff.ble function with $f(0) = f'(0) = 0$. If $|f''(x)| \leq 1 \forall x \in \mathbb{R}$, show that $|f(x)| \leq \frac{1}{2} \forall x \in [-1, 1]$

L' Hopital's Rule: Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a map. We know the meaning of $\lim_{x \rightarrow c} f(x) = L$ when $c, L \in \mathbb{R}$. what if L and/or c is either $\pm \infty$?

Defn: Let $L \in \mathbb{R}$ and $c = +\infty$. Then $\lim_{x \rightarrow \infty} f(x) = L$ means $\forall \epsilon > 0, \exists A = A(\epsilon) \in \mathbb{R}$ such that $x > A \Rightarrow |f(x) - L| < \epsilon$.
Similarly, one can define $\lim_{x \rightarrow -\infty} f(x) = L$

Defn: Let $c \in \mathbb{R}$ and $L = \infty$. Then $\lim_{x \rightarrow c} f(x) = \infty$ means given any $A \in \mathbb{R}, \exists \delta = \delta(A) \in (0, \infty)$ such that $|x - c| < \delta \Rightarrow f(x) > A$

Defn: Let $c = \infty, L = \infty$. Then $\lim_{x \rightarrow \infty} f(x) = \infty$ means given any $A \in \mathbb{R}, \exists B = B(A) \in \mathbb{R}$ such that $x > B \Rightarrow f(x) > A$

Suppose $c, L \in \mathbb{R} \cup \{-\infty, \infty\}$. This means that each of c and L is either a real no. or $\pm \infty$.

By a deleted nbhd $\mathcal{I}$ of c , we mean:

- $(c-\delta, c+\delta) \setminus \{c\}$ for some $\delta > 0$ if $c \in \mathbb{R}$
- (A, ∞) for some $A \in \mathbb{R}$ if $c = \infty$
- $(-\infty, B)$ for some $B \in \mathbb{R}$ if $c = -\infty$

Thm: (L'Hopital's Rule / Bernoulli's Rule)

Suppose f, g are two real-valued diff'ble functions defined on deleted nbhd $\mathcal{I}$ of $c \in \mathbb{R} \cup \{-\infty, \infty\}$. Assume that $g'(x) \neq 0 \forall x \in \mathcal{I}$. Also assume that

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} \in \{0, \infty, -\infty\},$$

and $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} = L$. Then,

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = L.$$

Riemann Integration

Fix $a < b$, $a, b \in \mathbb{R}$. Also fix a bounded function $f: [a, b] \rightarrow \mathbb{R}$, i.e., $\exists M \in (0, \infty)$ such that $|f(x)| \leq M \forall x \in [a, b]$.

$$\text{Area under curve} = \int_a^b f(x) dx$$

Defn: A partition P of $[a, b]$ is an ordered set of values $\{a = x_0 < x_1 < x_2 < \dots < x_n = b\}$ for some $n \in \mathbb{N}$.

$\|P\|$ = mesh of partition P

$$= \max_{1 \leq i \leq n} |x_i - x_{i-1}|$$

Defn: Suppose P_1 and P_2 are two partitions of $[a, b]$. We say that P_2 is a refinement of P_1 , or P_2 is a finer partition than P_1 , if as sets $P_1 \subseteq P_2$.

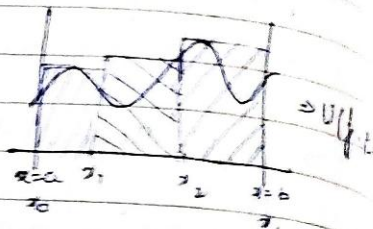
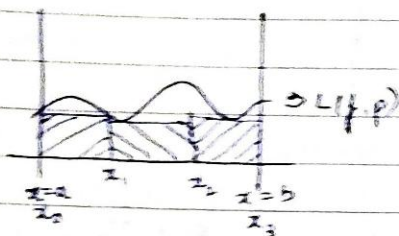
We say that P_1 is a coarser partition than P_2 .

Q. If $P_1 \subseteq P_2$ then $\|P_2\| \leq \|P_1\|$

Def: Given a partition p of $[a, b]$, we define

$$L(f, p) := \sum_{i=1}^n \left(\inf_{t \in [x_{i-1}, x_i]} f(t) \right) (x_i - x_{i-1})$$

= the lower sum of f w.r.t p .



$$U(f, p) := \sum_{i=1}^n \left(\sup_{t \in [x_{i-1}, x_i]} f(t) \right) (x_i - x_{i-1})$$

= The upper sum of f w.r.t p

Observation: $L(f, p) \leq U(f, p) \quad \forall$ partition p of $[a, b]$

Q: Take any partition $P = \{a = x_0 < x_1 < x_2 < \dots < x_n = b\}$
 Take points $t_1 \in [x_0, x_1], t_2 \in [x_1, x_2], \dots, t_n \in [x_{n-1}, x_n]$

Show that

$$\left| \sum_{i=1}^n f(t_i) (x_i - x_{i-1}) \right| \leq M(b-a), \quad \text{where } |f(x)| \leq M \quad \forall x \in [a, b]$$

If f is cont., then show that

$$-M(b-a) \leq L(f, p) \leq U(f, p) \leq M(b-a) \quad \forall \text{ partition } p \text{ of } [a, b]$$

In particular, the sets

$$\{L(f, p) : p \text{ is a partition of } [a, b]\}$$

$$\{U(f, p) : p \text{ is a partition of } [a, b]\}$$

are both non-empty and bounded.

Def: $\int_a^b f(x) dx := \sup \{L(f, p) : p \text{ is a partition of } [a, b]\}$
 = Lower Riemann integral of f on/over $[a, b]$

$\int_a^b f(x) dx := \inf \{U(f, p) : p \text{ is a partition of } [a, b]\}$
 = Upper Riemann integral of f on/over $[a, b]$

$$\int_a^b f(x) dx \leq \int_a^b f(x) dx$$

Defn: f is called Riemann integrable on $[a, b]$ if

$$\int_a^b f(x) dx = \int_a^b f(x) dx = I \text{ (say)}$$

In this case, we define I to be the Riemann Integral of f on/over $[a, b]$ and write

$$I = \int_a^b f(x) dx = \int_{[a, b]} f(x) dx$$

Notation $R[a, b]$ = the set of all Riemann Integrable functions on $[a, b]$.
 $f \in R[a, b]$ will mean f is Riemann Integrable on $[a, b]$.

Properties

$$f \in R[a, b] \Rightarrow |f| \in R[a, b]$$

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

Ex: $f: [0, 1] \rightarrow \mathbb{R}$ $f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$ isn't Riemann integrable

Thm: (i) If $f: [a, b] \rightarrow \mathbb{R}$ is a function with only finitely many discontinuities, then $f \in R[a, b]$. In particular, if f is continuous on $[a, b]$, it's Riemann Integrable.
 (ii) If $f \in R[a, b]$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then $g \circ f \in R[a, b]$

Defn: Suppose I is a (bdd or unbdd) open interval and $f: I \rightarrow \mathbb{R}$ is a map. A diffble fn. $F: I \rightarrow \mathbb{R}$ is called a primitive/anti-derivative of f on I if $F'(x) = f(x) \forall x \in I$

Notation $F(x) = \int f(x) dx$

→ Primitive of a function isn't unique

Ex: If F_1 and F_2 are two primitives of f on an open interval I , then $\exists c \in \mathbb{R}$ such that

$$F_1(x) = F_2(x) + c \quad \forall x \in I$$

Thm Suppose I is an open interval and $f: I \rightarrow \mathbb{R}$ is a continuous function. Let $F: I \rightarrow \mathbb{R}$ be a primitive of f on I . Then $\forall c, x \in I$,

$$F(x) = F(c) + \int_c^x f(t) dt$$

i.e.,
$$\int_c^x f(t) dt = F(x) - F(c)$$

[We have defined $\int_a^b f(x) dx$ when $a < b$ and $f \in R[a, b]$.
 $\int_a^a f(x) dx := 0$

$$\int_b^a f(x) dx := - \int_a^b f(x) dx$$

Thm (First Fundamental Thm of Calculus)

Suppose $f \in R[a, b]$. Fix $c \in [a, b]$ and define $F: [a, b] \rightarrow \mathbb{R}$ by $F(x) = \int_c^x f(t) dt$, $x \in [a, b]$

Then we have: F is cont. on $[a, b]$.

$\rightarrow$ If $x_0 \in (a, b)$ is a point of continuity of f , then F is differentiable at x_0 and $F'(x_0) = f(x_0)$

Q. Prove the second thm. from the first thm.

Sketch i. $f \in R[a, b] \Rightarrow f$ bdd.

Suppose $|f(x)| \leq M \quad \forall x \in [a, b]$

To Show: $F: [a, b] \rightarrow \mathbb{R}$ defined by

$$F(x) = \int_c^x f(t) dt, \quad x \in [a, b] \quad (c \in [a, b] \text{ is fixed})$$

is continuous.

Fix $x_0 \in [a, b]$. To Show: F is cont. at x_0 .

Taking any $x \in [a, b]$, we look at $|F(x) - F(x_0)|$

Recall: $\forall a \leq b, g \in R[a, b]$,

$$\left| \int_a^b g(t) dt \right| \leq \int_a^b |g(t)| dt$$

$$\begin{aligned} |F(x) - F(x_0)| &= \left| \int_c^x f(t) dt - \int_c^{x_0} f(t) dt \right| = \left| \int_{x_0}^x f(t) dt \right| \\ &= \begin{cases} \int_{x_0}^x |f(t)| dt & x \geq x_0 \\ \int_x^{x_0} |f(t)| dt & x < x_0 \end{cases} \end{aligned}$$

$$\leq \begin{cases} M(x-x_0) & \text{if } x \geq x_0 \\ M(x_0-x) & \text{if } x < x_0 \end{cases}$$

$$= M|x-x_0|$$

$$\Rightarrow \forall x \in [a, b], |F(x) - F(x_0)| \leq M|x-x_0| \quad - (1)$$

Q. Using (1), Show that F is cont. at x_0

→ Take $x_0 \in (a, b)$ such that f is cont. at x_0 .

To show: F is diffble at x_0 and $F'(x_0) = f(x_0)$. Take h

small enough so that $x_0+h \in (a, b)$

$$F(x_0+h) - F(x_0) = \int_{x_0}^{x_0+h} f(t) dt$$

$$= h f(x_0) + \int_{x_0}^{x_0+h} (f(t) - f(x_0)) dt$$

$$\frac{F(x_0+h) - F(x_0)}{h} = f(x_0) + \frac{1}{h} \int_{x_0}^{x_0+h} (f(t) - f(x_0)) dt$$

→ If we show $\lim_{h \rightarrow 0} G(h) = 0$, then $G(h)$

$$\lim_{h \rightarrow 0} \frac{F(x_0+h) - F(x_0)}{h} = f(x_0) \quad \text{proving } F \text{ is diffble at } x_0 \text{ and}$$

$$F'(x_0) = f(x_0)$$

Using continuity of f at x_0 , it can be shown that
 $\forall \epsilon > 0 \quad \exists \delta = \delta(\epsilon) > 0$ such that $0 < |h| < \delta \Rightarrow$

$$|G(h)| < \epsilon \quad \text{establishing} \quad \lim_{h \rightarrow 0} G(h) = 0$$

Thm: Assume that g has a const. derivative g' on an open interval I . Let $J = \{g(x) : x \in I\}$.

Assume $f: J \rightarrow \mathbb{R}$ is another fn. which is cont. J .

Then $\forall x \in I$ & $c \in I$, we have

$$\int_c^x f(g(t)) g'(t) dt = \int_{g(c)}^{g(x)} f(u) du, \quad u = g(t)$$

thm: Integration by Parts

Suppose $f: [a, b] \rightarrow \mathbb{R}$ and $g: [a, b] \rightarrow \mathbb{R}$ are two continuously diffble functions. Then

$$\int_a^b f(x) g'(x) dx = (f(b)g(b) - f(a)g(a)) - \int_a^b f'(x)g(x) dx$$

Improper Riemann Integral

- ① $\rightarrow$ Define Riemann Integral on unbdd interval $\rightarrow$ Improper Riemann Integral of first kind.
- ② $\rightarrow$ Define Riemann Integral for unbdd fn. on bdd interval $\rightarrow$ Improper Riemann Integral of second kind

- ① Fix $a \in \mathbb{R}$. Suppose $f: [a, \infty) \rightarrow \mathbb{R}$ is a bdd function such that $f \in R[a, b] \forall b > a$. If further $\lim_{b \rightarrow \infty} \int_a^b f(x) dx = I \in \mathbb{R}$ exists and is finite, then

we say that f is Riemann Integrable on $[a, \infty)$ and define improper integral as

$$\int_a^\infty f(x) dx := \lim_{b \rightarrow \infty} \int_a^b f(x) dx = I$$

In this case we also say that the integral $\int_a^\infty f(x) dx$ exists/converges. On the other hand, if $f \geq 0$ and $f \in R[a, b] \forall b > a$ but $\lim_{b \rightarrow \infty} \int_a^b f(x) dx = \infty$, then we say that $\int_a^\infty f(x) dx$ diverges.

Similarly, fixing $b \in \mathbb{R}$ and taking limit (if exists and is finite) as $a \rightarrow -\infty$, we can def. R integ. on $(-\infty, b]$ and

$$\int_{-\infty}^b f(x) dx := \lim_{a \rightarrow -\infty} \int_a^b f(x) dx, \text{ provided lim exist and is finite.}$$

$\rightarrow$ Let $f \in R(-\infty, c] \cap [c, \infty)$

$$\int_{-\infty}^\infty f(x) dx = \int_{\mathbb{R}} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx$$

$$\int_1^{\infty} x^{-s} dx \begin{cases} < \infty & \text{if } s > 1 \\ = \infty & \text{if } s \leq 1 \end{cases}$$

Thm: Comparison Test

Suppose $a \in \mathbb{R}$ and $f: [a, \infty) \rightarrow [0, \infty)$ and $g: [a, \infty) \rightarrow [0, \infty)$ are two maps. Suppose that $f, g \in R[a, b] \forall b > a$ and $0 \leq f(x) \leq g(x) \forall x \geq a$.

Then we have:

i) If $\int_a^{\infty} g(x) dx < \infty$, then $\int_a^{\infty} f(x) dx < \infty$

ii) If $\int_a^{\infty} f(x) dx = \infty$, then $\int_a^{\infty} g(x) dx = \infty$

Thm: Ratio Test

Suppose $a \in \mathbb{R}$, and $f: [a, \infty) \rightarrow [0, \infty)$ and $g: [a, \infty) \rightarrow [0, \infty)$ are two functions such that $f, g \in R[a, b] \forall b > a$

if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = c \in (0, \infty)$, then

either both $\int_a^{\infty} f(x) dx$, $\int_a^{\infty} g(x) dx$ converge or both diverge

→ Moreover, if $c = 0$, then

$$\int_a^{\infty} g(x) dx < \infty \Rightarrow \int_a^{\infty} f(x) dx < \infty$$

for any $s \in \mathbb{R}$ $\int_1^{\infty} x^s e^{-x} dx < \infty$

Proof: Note that $\lim_{x \rightarrow \infty} \frac{x^s e^{-x}}{x^{-2}} = \lim_{x \rightarrow \infty} \frac{x^{s+2}}{e^x} = 0$

Hence $\exists A > 1$ s.t. $x > A \Rightarrow \frac{x^s e^{-x}}{x^{-2}} < 1$

i.e., $x^s e^{-x} < x^{-2}$

We know $\int_A^{\infty} x^{-2} dx \leq \int_1^{\infty} x^{-2} dx < \infty$

By comparison test $\Rightarrow \int_A^{\infty} x^s e^{-x} dx < \infty$

$$\Rightarrow \int_1^{\infty} x^s e^{-x} dx < \infty$$

Q. $\int_1^{\infty} \frac{dx}{x^2+x^3+x^4} < \infty$

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x^2+x^3+x^4}}{\frac{1}{x^4}} = \lim_{x \rightarrow \infty} \frac{x^4}{x^4+x^3+x^2} = 1 \in (0, \infty)$$

Also, $\int_1^{\infty} \frac{dx}{x^4} < \infty$

$\therefore$ By ratio test, $\int_1^{\infty} \frac{dx}{x^2+x^3+x^4} < \infty$

Q. check if the following integrals converge -

i. $\int_1^{\infty} \frac{dx}{\sqrt{x+x^2}}$

ii. $\int_1^{\infty} \frac{dx}{\sqrt{x^2+4x}}$

iii. $\int_1^{\infty} \frac{e^{-x}}{\sqrt{x+x^2}} dx$

Seq. of functions: Defn and Examples

If for each $n \in \mathbb{N}$, $f_n: S \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is a function, then we call $\{f_n\}_{n \in \mathbb{N}}$ a sequence of functions defined on S .

Ex: $S = [0, 1]$. $f_n = x^n$, $x \in [0, 1]$ $\forall n \in \mathbb{N}$.

$S = \mathbb{R}$. $f_n(x) = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$, $x \in \mathbb{R}$.

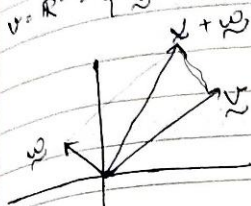
Linear Algebra

Page: / /

Date: / /

Recall that $\mathbb{R}$ satisfies (A1) - (A6). Note that $\mathbb{C}$ also satisfies (A1) - (A6).

$$V = \mathbb{R}^2 = \{ \underline{v} = (v_1, v_2) : v_1 \in \mathbb{R}, v_2 \in \mathbb{R} \}$$



$$\underline{u} + \underline{w} = (u_1 + w_1, u_2 + w_2)$$

$$\forall \alpha \in \mathbb{R} \text{ and } \forall \underline{u} \in V = \mathbb{R}^2, \\ \alpha \underline{u} = \alpha(x_1, x_2) = (\alpha x_1, \alpha x_2)$$

Note: we have a binary operation

$$V \times V \rightarrow V$$

$$(\underline{u}, \underline{w}) \mapsto (\underline{u} + \underline{w}) \rightarrow \text{Vector addition}$$

find

Another map $\mathbb{R} \times V \rightarrow V$

$$(\alpha, \underline{u}) \mapsto \alpha \underline{u} \rightarrow \text{Scalar Multiplication}$$

Associativity $\forall \underline{u}, \underline{v}, \underline{w} \in V,$
 $(\underline{u} + \underline{v}) + \underline{w} = \underline{u} + (\underline{v} + \underline{w})$

Existence of Inverse Elements for Vector Addition

For each $\underline{u} \in V, \exists -\underline{u} \in V$ such that

$$\underline{u} + (-\underline{u}) = \underline{0}$$

$\forall \alpha, \beta \in \mathbb{R}$ and $\forall \underline{u} \in V,$

$$\alpha(\beta \underline{u}) = (\alpha\beta) \underline{u}$$

$\forall \alpha \in \mathbb{R}$ and $\forall \underline{u}, \underline{v} \in V,$

$$\alpha(\underline{u} + \underline{v}) = \alpha \underline{u} + \alpha \underline{v}$$

$$(\alpha + \beta) \underline{u} = \alpha \underline{u} + \beta \underline{u}$$

$$1 \underline{u} = \underline{u} \rightarrow \text{Multiplicative Identity}$$

Fix $n \in \mathbb{N}$.

$V = \mathbb{R}^n = \{ \underline{v} = (v_1, v_2, \dots, v_n) \}$ and define vector addition
 $\underline{v} + \underline{w} := (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n)$, where
 $\underline{v} = (v_1, v_2, \dots, v_n)$, $\underline{w} = (w_1, w_2, \dots, w_n) \in V = \mathbb{R}^n$

Also define scalar multiplication

$$\alpha \underline{v} = \alpha (v_1, v_2, \dots, v_n) := (\alpha v_1, \alpha v_2, \dots, \alpha v_n)$$

where $\alpha \in \mathbb{R}$, $\underline{v} \in V = \mathbb{R}^n$.

Ex: $X = [0, 1]$ and look at the space.

$V = \mathbb{R}^X = \{ f: X \rightarrow \mathbb{R} \text{ is a map} \}$ of all real valued functions defined on $X = [0, 1]$

Define: Vector addition and scalar multiplication pointwise, i.e., $\forall \alpha \in \mathbb{R}$ and $\forall f, g \in V$, define $f + g \in V$ and $\alpha f \in V$ as follows:

$$\begin{aligned} (f + g)(x) &= f(x) + g(x), & x \in X \\ (\alpha f)(x) &= \alpha f(x), & x \in X \end{aligned}$$

It can be checked that V also satisfies (VS1) - (VS8) with the zero function $\underline{0}$ (i.e., $\underline{0}(x) = 0 \forall x \in X$) being the identity element for vector addition and negative of a function $f \in V$ (i.e., $(-f)(x) := -f(x) \forall x \in X$) being the vector additive inverse of f .

$$S = \{ (v_1, v_2) \in V = \mathbb{R}^2 : v_1 = v_2 \} \subseteq V = \mathbb{R}^2$$

Let S borrow vector addition and scalar multiplication from $V = \mathbb{R}^2$. One can check that S also satisfies (VS1) - (VS8) with these borrowed operations.

Defn: Let V be a non-empty set with a binary operation called vector addition on V , i.e., a map

$$\begin{aligned} V \times V &\rightarrow V \text{ defined by} \\ (\underline{v}, \underline{w}) &\mapsto \underline{v} + \underline{w} \end{aligned}$$

and a scalar multiplication by real numbers i.e., a map $\mathbb{R} \times V \rightarrow V$ defined by

$(\alpha, \underline{v}) \mapsto \alpha \underline{v} \quad \forall \underline{v} \in V$
such that the axioms (VS1) - (VS8) are satisfied.
Then we say that V is a vector space/linear space over $\mathbb{R}$.

Remarks: Elements of $V \rightarrow$ vectors
" " $\mathbb{R} \rightarrow$ scalars

$\rightarrow$ vector space are generally over any field F (generally $\mathbb{R}$)

Thm: Suppose V is a vector space over $\mathbb{R}$. Then

$\forall \underline{u}, \underline{v}, \underline{w} \in V$ and $\forall \alpha, \beta, \gamma \in \mathbb{R}$, we have
i. $\underline{u} + \underline{v} = \underline{u} + \underline{w} \Rightarrow \underline{v} = \underline{w}$ (Cancellation Law)

ii. $\underline{0}$ is unique

iii. $-\underline{v}$ is unique.

iv. $\underline{v} + \underline{v} = \underline{v} \Rightarrow \underline{v} = \underline{0}$

v. $0 \underline{v} = \underline{0}$

vi. $\alpha \underline{0} = \underline{0}$

vii. $\alpha \underline{v} = \underline{0} \Rightarrow \alpha = 0$ or $\underline{v} = \underline{0}$

(Cancellation Law for Scalar Mult.)

viii. $(-\alpha) \underline{v} = -(\alpha \underline{v})$

S. $\alpha + \beta$

- If V is a vector space over $\mathbb{R}$ and $\emptyset \neq S \subseteq V$, then S is called a vector subspace of V if S , with vector addition and scalar multiplication borrowed from V , forms a vector space.

Ex. $V = \mathbb{R}^2$

$S = \{ \underline{v} = (u, v) \in \mathbb{R}^2, u, v = 1 \}$

Then, S is a linear subspace of V .

Let $X = [0, 1]$

$V = \mathbb{R}^X = \{ f : X \rightarrow \mathbb{R} \text{ is a map} \}$

Define $S = \{ f \in V : f \text{ is cont.} \}$

$= \{ f : [0, 1] \rightarrow \mathbb{R} \text{ is a cont. map} \}$

Show that S is a vector subspace of V .

Remark: If we define $S_1 = \mathbb{R}[0,1]$, then it can be shown that S_1 is also a linear subspace of V .

$$\underset{\text{(Subsp.)}}{S} \subseteq \underset{\text{(Subsp.)}}{S_1} \subseteq V$$

Thm: Let V be any vector space. A nonempty subset $S \subseteq V$ is a subspace of V iff S is closed under taking linear combinations i.e., $\alpha \underline{u} + \beta \underline{v} \in S$ for $\alpha, \beta \in \mathbb{R}$ and $\underline{u}, \underline{v} \in S$.

Defn: Let V be a vector space (over $\mathbb{R}$). We take $k \in \mathbb{N}$ and vectors $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k \in V$. Any vector of the form $\underline{v} = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \dots + \alpha_k \underline{v}_k = \sum_{i=1}^k \alpha_i \underline{v}_i$ where $\alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{R}$, is called a linear combination of $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k \in V$.

Ex $\rightarrow V = \mathbb{R}^2$ and $B = \{ \underline{e}_1 := (1,0), \underline{e}_2 := (0,1) \} \subseteq V$
 any $\underline{v} \in V = \mathbb{R}^2$ can be written as

$$\begin{aligned} \underline{v} &= (v_1, v_2) = (v_1, 0) + (0, v_2) \\ &= v_1 (1,0) + v_2 (0,1) \\ &= v_1 \underline{e}_1 + v_2 \underline{e}_2 \end{aligned}$$

We shall say that V is the "linear span" $\bar{B} = \text{sp}(B)$ of B .

Defn: Let V be any vector space. If a non-empty set $B = \{ \underline{v}_1, \underline{v}_2, \dots, \underline{v}_k \} \subseteq V$ of vectors, we define the linear span of B as the set

$\bar{B} = \text{sp}(B) := \left\{ \sum_{i=1}^k \alpha_i \underline{v}_i : \alpha_i \in \mathbb{R} \right\}$ of all possible linear combination of elements of B . We also define

$$\bar{\emptyset} = \text{sp}(\emptyset) = \{ \underline{0} \}$$

Q. Let V be a vector space. $B \subseteq V$ is any finite subset. Show that $\bar{B} \subseteq V$.

$\rightarrow$ In the above example, $V = \mathbb{R}^2$, $B = \{ \underline{e}_1 = (1,0) \text{ and } \underline{e}_2 = (0,1) \}$ and $\bar{B} = V = \mathbb{R}^2$

Def: In the above setup, whenever $S = \overline{B} = \{v_1, v_2, \dots, v_n\}$ we say that vectors v_i generate S or say B generates S .
 More generally, if $V = \mathbb{R}^n$, then $B = \{e_1, e_2, \dots, e_n\}$ where
 $e_1 = (1, 0, \dots, 0) \in \mathbb{R}^n$
 $e_2 = (0, 1, 0, \dots, 0) \in \mathbb{R}^n$
 $e_n = (0, 0, \dots, 0, 1) \in \mathbb{R}^n$ generates V .

$$\begin{aligned} v &= (v_1, v_2, \dots, v_n) \\ &= (v_1, 0, \dots, 0) + (0, v_2, \dots, 0) + \dots + (0, 0, \dots, v_n) \\ &= \sum v_i e_i \end{aligned}$$

$$B_1 = \{e_1, e_2, e_1 + e_2\} = \{(1, 0), (0, 1), (1, 1)\} \text{ also generates } V.$$

Proof: Clearly $\overline{B_1} \subseteq V$.

To show $V \subseteq \overline{B_1}$

Take any vector $v = (v_1, v_2) \in V = \mathbb{R}^2$

$$\text{Then } v = v_1 e_1 + v_2 e_2 + 0(e_1 + e_2) \in \overline{B_1}$$

However $(1, 1) = e_1 + e_2$ is redundant because e_1 and e_2 generate V anyway.

Def: For any vector space V , a finite collection $C = \{v_1, v_2, \dots, v_k\} (\neq \emptyset) \subseteq V$ of vector is called

linearly independent if
 $\sum_{i=1}^k \alpha_i v_i = 0, \alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{R}$
 $\Rightarrow \alpha_i = 0 \quad \forall i$

$\rightarrow$ In other words, no nontrivial linear combination of

$v_1, v_2, \dots, v_k$ is equal to 0 .

Ex: $V = \mathbb{R}^2, e_1, e_2, e_1 + e_2$ are not linearly independent.

② In the setup of ①, if $\exists \alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{R}$ s.t.
 $(\alpha_1, \alpha_2, \dots, \alpha_k) \neq (0, 0, \dots, 0)$

and $\sum \alpha_i v_i = 0$, then

v_i are called linearly dependent.

Let V be any vector space. Then we have:

- i. If $B(\subseteq V)$ generates $S \subseteq V$, then so does any superset of B .
- ii. If $C(\subseteq V)$ is linearly independent, then so is any subset of C .
- iii. If $0 \in C$, then C is always linear dep.
- iv. $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k \in V$ are linearly dependent iff at least one of these vectors can be written as a linear combination of other vectors. In particular, for $k=2$, $\underline{v}_1$ and $\underline{v}_2$ are lin. dependent iff one of them is scalar multiple of other.

Proof: If part: Suppose without loss of generality (WLOG) that $\underline{v}_1 = \sum_{i=2}^k \alpha_i \underline{v}_i$, where $\alpha_i \in \mathbb{R} \forall i$. Then

$$1 \cdot \underline{v}_1 + (-\alpha_2) \underline{v}_2 + (-\alpha_3) \underline{v}_3 + \dots + (-\alpha_k) \underline{v}_k = \underline{0}$$

only if part

Suppose $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k$ are lin. dep.

$$\sum \alpha_i \underline{v}_i = \underline{0} \quad \text{for some } \alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{R}$$

Let $\alpha_j \neq 0$. Then

$$\alpha_j \underline{v}_j = \sum_{\substack{i=1 \\ i \neq j}}^k (-\alpha_i) \underline{v}_i$$

$$\underline{v}_j = \sum_{\substack{i=1 \\ i \neq j}}^k \left(\frac{-\alpha_i}{\alpha_j} \right) \underline{v}_i$$

$\Rightarrow \underline{v}_j$ is a lin. comb. of the rest.

Note that $B = \{ \underline{e}_1, \underline{e}_2, \dots, \underline{e}_n \} \subseteq V = \mathbb{R}^n$ is a linearly independent set that generates $V = \mathbb{R}^n$.

Defⁿ: Let V be any vector space over $\mathbb{R}$.

If $k \in \mathbb{N} \cup \{0\}$, $B = \{ \underline{v}_1, \underline{v}_2, \dots, \underline{v}_k \}$.
 $= \emptyset$ if $k=0$

$\subseteq V$ (finite subset) is linear ind. such that

$\overline{B} = V$, then we say that V is a finite dimensional vector space over $\mathbb{R}$ and B is called a basis of V .

Thm: Suppose V is a finite dimensional vector space over $\mathbb{R}$ (f.d.v.s). $B \subseteq V$ (finite). The following are equivalent (TFAE) :-

- i. $B = \{v_1, v_2, \dots, v_k\}$ is a basis for V .
- ii. B is a minimal generating set for V , i.e., no proper subset of B generates V .
- iii. B is a maximal linearly independent subset of V , i.e., $\forall v \in V \setminus B$, $B \cup \{v\}$ is linearly dependent.
- iv. Each vector $w \in V$ can be written uniquely as a linear combination of $v_1, v_2, \dots, v_k$ i.e., $\forall w \in V$, $\exists$ unique $\alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{R}$ s.t., $w = \sum_{i=1}^k \alpha_i v_i$

Ex - $V = \mathbb{R}^n$

8. Show that following sets are all bases of V .
- $B = \{e_i\} = \text{std. basis of } \mathbb{R}^n$
- $B_1 = \{e_1, e_1 + e_2, e_1 + e_2 + e_3, \dots, e_1 + e_2 + \dots + e_n\}$
- $B_2 = \{e_1, e_1 + e_2, e_3, \dots, e_n\}$

Thm: Let V be a f.d.v.s. Then any two bases for V has the same size.

Defn: Let V be a f.d.v.s and B be any basis for V . Then we define the dimension of V as $\dim(V) := |B|$

From the ex, it follows that

$$\dim(\mathbb{R}^n) = n \quad \forall n \in \mathbb{N}.$$

On the other hand $\dim(\{0\}) = 0$

$\therefore$ It is customary to define

$$\mathbb{R}^0 := \{0\}.$$

Ex - $V = \mathbb{R}^3$

$$S = \{(v_1, v_2, v_3) \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 0\}$$

It is possible to check that $S \subseteq V$

$$S = \{(v_1, v_2, v_3) \in \mathbb{R}^3 : v_3 = -v_1 - v_2\}$$

Page: _____
Date: ____/____/____

$$S = \{ (v_1, v_2, -v_1 - v_2) : v_1, v_2 \in \mathbb{R} \}$$

$$\Rightarrow \text{any } \underline{v} = (v_1, v_2, v_3) \in S \text{ is of the form}$$

$$\underline{v} = (v_1, v_2, -v_1 - v_2)$$

$$= (v_1, 0, -v_1) + (0, v_2, -v_2)$$

$$= v_1(1, 0, -1) + v_2(0, 1, -1)$$

Claim: $B = \{(1, 0, -1), (0, 1, -1)\}$ is a basis for S .

Proof: We just showed $V = \overline{B}$

Take, $\alpha_1, \alpha_2 \in \mathbb{R}$ such that

$$\alpha_1(1, 0, -1) + \alpha_2(0, 1, -1) = (0, 0, 0)$$

$$\Rightarrow (\alpha_1, \alpha_2, -\alpha_1 - \alpha_2) = (0, 0, 0)$$

$$\Rightarrow \alpha_1 = 0 \text{ and } \alpha_2 = 0$$

$\Rightarrow B$ is lin. independent and $\overline{B} = S$

B is a basis for S .

In part, $\dim(S) = |B| = 2$

Q. $S_1 = \{ \underline{v} = (v_1, v_2, v_3) \in \mathbb{R}^3 : v_1 = v_2 = v_3 \}$

$$S_2 = \{ \underline{v} = (v_1, v_2, v_3) \in \mathbb{R}^3 : v_1 = v_2 \}$$

Show that $S_1 \subseteq S_2 \subseteq V$

Find a basis for S_1 and show that $\dim(S_1) = 1$

Find a basis for S_2 and show that $\dim(S_2) = 2$

Defn: Let V be a f.d.v.s and S_1, S_2 are two subspaces of V . Define

$$S_1 + S_2 := \{ \underline{v}_1 + \underline{v}_2 : \underline{v}_1 \in S_1, \underline{v}_2 \in S_2 \}$$

Suppose $V = \mathbb{R}^2$. Let $S \subseteq V$.

$$\Rightarrow \dim(S) \leq \dim(V) = 2$$

$$\dim(S) \in \{0, 1, 2\}$$

If $S = \{ \underline{0} \}$, then $\dim(S) = 0$

If $S = V = \mathbb{R}^2$, then $\dim(S) = 2$

For all other subspaces, $\dim(S) = 1$

Suppose $\dim(S) = 1$

Take any $\underline{v} \in S \setminus \{ \underline{0} \}$.

Then it can be shown that

$$S = \{ \underline{v} \}. \text{ This means that}$$

$$S = \{ \alpha \underline{v} = \alpha \underline{v}_1 + \alpha \underline{v}_2 : \alpha \in \mathbb{R} \}$$

$\dim(S) = 1 \Rightarrow S$ is a 1D line passing through $\underline{0}$.

Take $S_1 = \{ (v_1, v_2) \in \mathbb{R}^2 : v_1 = 0 \} = \{ (0, v_2) : v_2 \in \mathbb{R} \}$
 $S_2 = \{ (v_1, v_2) \in \mathbb{R}^2 : v_2 = 0 \}$

clearly $S_1 \cap S_2 = \{ \underline{0} \} = \{ (0, 0) \}$
 $S_1 + S_2 = V = \mathbb{R}^2$

Any vector $\underline{v} = (v_1, v_2)$ can be written as
 $\underline{v} = (v_1, 0) + (0, v_2)$

Thm: Let V be f.d.v.s with $\dim(V) = n \geq 2$

Thm: Let V be a f.d.v.s and $S_1, S_2 \subseteq V$,
 $\dim(S_1 + S_2) = \dim(S_1) + \dim(S_2) - \dim(S_1 \cap S_2)$.

Sketch: $\dim(S_1 \cap S_2) = l$ $\dim(S_1) = k + l$
 $\dim(S_2) = l + m$, $\dim(S_1 + S_2) = k + l + m$

Application Let $V = \mathbb{R}^2$

$S_1 = \{ (v_1, v_2) \in \mathbb{R}^2 : v_1 - v_2 = 0 \}$

$S_2 = \{ (v_1, v_2) \in \mathbb{R}^2 : v_1 + v_2 = 0 \}$

$S_1 \subseteq V, S_2 \subseteq V$

$S_1 \cap S_2 = \{ \underline{0} \}$

$\dim(S_1) = \dim(S_2) = 1$

$\dim(S_1 + S_2) = \dim(S_1) + \dim(S_2) - \dim(S_1 \cap S_2) = 2$

$\Rightarrow S_1 + S_2 = V = \mathbb{R}^2$

Linear Map / Linear Transformation

Let V_1, V_2 be f.d.v.s

A map $T: V_1 \rightarrow V_2$ is called a lin. map / transformation

$\rightarrow$ if $\forall \alpha, \beta \in \mathbb{R}, \forall \underline{u}, \underline{v} \in V_1$,
 $T(\alpha \underline{u} + \beta \underline{v}) = \alpha T(\underline{u}) + \beta T(\underline{v})$

$\rightarrow$ if $V_1 = V_2 = V$ (say) and $T: V \rightarrow V$ is a linear map, then

T is called a linear operator

$\rightarrow$ if $V = \mathbb{R}$ (also a f.d.v.s over $\mathbb{R}$, $\dim(V) = 1$)

Let $T: V_1 \rightarrow V_2$ be a linear map. V_1 and V_2 are vector spaces over $\mathbb{R}$. T is called a linear transformation. Notation: $T: V_1 \rightarrow V_2$.

Thm: If $T: V_1 \rightarrow V_2$ is a linear map, then $\dim(V_1) = \dim(V_2)$.

If $\dim(V_1) = n \in \mathbb{R}^n$, then $V_2 = \mathbb{R}^n$.
 $V_1 = \mathbb{R}^2, V_2 = \mathbb{R}$

Define $T: V_1 \rightarrow V_2$ by

$$T(u, v) = 2u - 3v \quad \forall (u, v) \in V_1$$

Then T is a linear function because $\forall \alpha, \beta \in \mathbb{R}$ and $u, v \in V_1 \subset \mathbb{R}^2$

$$\begin{aligned} T(\alpha u + \beta v) &= T(\alpha(u, v) + \beta(v, v)) \\ &= T(\alpha u + \beta v, \alpha v + \beta v) \\ &= 2(\alpha u + \beta v) - 3(\alpha v + \beta v) \\ &= 2\alpha u + 2\beta v - 3\alpha v - 3\beta v \\ &= 2\alpha u - \alpha v - \beta v \\ &= \alpha(2u - v) - \beta v \\ &= \alpha T(u, v) - \beta T(v, v) \\ &= \alpha T(u, v) - \beta T(v, v) \end{aligned}$$

Q. $V_1 = \mathbb{R}^2, V_2 = \mathbb{R}^2, T: V_1 \rightarrow V_2$

Show that $T(u, v, w) = (u - v, v - w)$, $(u, v, w) \in \mathbb{R}^3$ is a linear transformation from V_1 to V_2 .

Q. $V_1 = V_2 = \mathbb{R}^2$

Show that $T: V_1 \rightarrow V_2$ defined by

$$T(u, v) = (u + v, u - v) \quad \forall (u, v) \in \mathbb{R}^2$$

is a linear operator on $\mathbb{R}^2$.

Q. Take $V_1 = \{ (u_1, u_2, u_3) : \sum u_i = 0 \} \subset \mathbb{R}^3$

$$V_2 = \{ (u_1, u_2) : \sum u_i = 0 \} \subset \mathbb{R}^2$$

$$V_3 = \mathbb{R}^2$$

Show the following:

$\rightarrow T_1: V_1 \rightarrow V_2$ defined by

$$T_1(u_1, u_2, u_3) = (u_1 + u_2, u_3) \text{ is a linear map from } V_1 \text{ to } V_2$$

$\rightarrow T_2: V_1 \rightarrow V_3$ defined by

$$T_2(u_1, u_2, u_3) = (u_1, u_2) \text{ is a vector space isomorphism}$$

from V to V

We know that any finite dimensional vector space over $\mathbb{R}$ looks like $\mathbb{R}^n$, where $n \in \mathbb{N}$.

Fix $m, n \in \mathbb{N}$.

What are all possible linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$?

From now on, whenever we write a vector (u, v, w) , it's always a column vector. Ex-

$$x \in \mathbb{R}^n \Rightarrow x = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}_{n \times 1}, \text{ a column vector in } \mathbb{R}^n.$$

To write a row vector, we use a transpose notation:
 $u^T = (u_1, u_2, \dots, u_n)_{1 \times n}$, a row vector in $\mathbb{R}^n$

Suppose $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map.

Let us denote by $e_1^{(n)}, e_2^{(n)}, \dots, e_n^{(n)}$ the standard basis elements for $\mathbb{R}^n$. $e_i^{(n)}$ are column vectors.

$e_1^{(m)}, e_2^{(m)}, \dots, e_m^{(m)}$ the standard basis elements for $\mathbb{R}^m$.

For each $j \in \{1, 2, \dots, n\}$,

$$T(e_j^{(n)}) \in \mathbb{R}^m = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix} \quad (\text{say})$$

$$= \sum_{i=1}^m a_{ij} e_i^{(m)}$$

Now take any $x \in \mathbb{R}^n$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \sum_{j=1}^n x_j e_j^{(n)}$$

Since $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map, we have

$$T(x) = T\left(\sum_{j=1}^n x_j e_j^{(n)}\right) = \sum_{j=1}^n x_j T(e_j^{(n)}) = \sum_{j=1}^n x_j \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{j=1}^n a_{1j} x_j \\ \sum_{j=1}^n a_{2j} x_j \\ \vdots \\ \sum_{j=1}^n a_{mj} x_j \end{pmatrix}_{m \times 1} \quad \leftarrow \text{column vector}$$

$$= \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = A x$$

$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map then, $\forall x \in \mathbb{R}^n$,
 $T(x) = Ax$ for some $m \times n$ matrix A .

Conversely, for any $m \times n$ matrix A with real entries, the map $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T(x) = Ax$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. This is because $\forall x, y \in \mathbb{R}^n$ and $\forall \alpha, \beta \in \mathbb{R}$.

$$A(\alpha x + \beta y) = \alpha Ax + \beta Ay$$

We have thus proved:

Thm: Fix $m, n \in \mathbb{N}$. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transform.

iff $\exists$ a $m \times n$ matrix A with real entries s.t.
 $T(x) = Ax \quad \forall x \in \mathbb{R}^n$

A is unique.

Defn: A is called the matrix for the linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ with respect to the standard bases.

Ex - ① If $A = 0$, then

$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the zero map, i.e., $T(x) = 0 \quad \forall x \in \mathbb{R}^n$

② If $m = n$ and $A = I_n$ (i.e., $a_{ij} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$)

$T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the identity map

$$T(x) = x \quad \forall x \in \mathbb{R}^n$$

③ $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T(x) = \begin{pmatrix} x_1 + x_2 \\ x_2 + x_3 \end{pmatrix} \quad \forall x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3$

T is a linear map.

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}_{2 \times 3}$$

$$T(e_1^{(3)}) = T\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1^{\text{st}} \text{ column}$$

$$T(e_2^{(3)}) = T\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2^{\text{nd}} \text{ column}$$

$$T(e_3^{(3)}) = T\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 3^{\text{rd}} \text{ column}$$

Q Show that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by
 $T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 + 2x_2 \\ 2x_1 - x_2 \\ x_1 + x_2 \end{pmatrix} \quad \forall x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2$ is a linear map.

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \\ 1 & 1 \end{pmatrix}_{3 \times 2}$$

Rank of a Matrix

Consider A is a matrix (with real entries)

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Consider by $A_{11}, A_{12}, \dots, A_{1n}$ as the $1^{st}, 2^{nd}, \dots, n^{th}$ row of A

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}$$

Consider by $A_{11}, A_{12}, \dots, A_{1n}$ as the $1^{st}, 2^{nd}, \dots, n^{th}$ column of A

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}$$

for $A_{11}, A_{12}, \dots, A_{1n} \in \mathbb{R}^m$

Column Space: $C(A) := \text{sp}\{A_{11}, A_{12}, \dots, A_{1n}\} \subseteq \mathbb{R}^m$
(Subspace)

Row Space: $R(A) := \text{sp}\{A_{11}, A_{12}, \dots, A_{1n}\} \subseteq \mathbb{R}^n$
(Subspace)

$$\dim(C(A)) = \dim(R(A))$$

column rank of A row rank of A

max. no. of linearly independent columns of A = max. no. of linearly independent rows of A

Definition The above dimensions are called the rank of a matrix A .
Notations: $\text{rank}(A) = \rho(A) = \dim(C(A)) = \dim(R(A))$

Remark: Rank plays an important role in prob/statistics/ data science

Clearly $\rho(A) \leq m$ Also $\rho(A) \leq n$
 $\rho(A) \leq \min\{m, n\}$

If $\rho(A) = \min\{m, n\}$, we call A a full rank matrix

Thm: Suppose $m = n$ i.e., A is a $n \times n$ matrix.
Then $\rho(A) = n$ iff A is invertible/nonsingular i.e., $\exists$ another matrix B s.t. $AB = BA = I_n$; $B = A^{-1}$
If $B = A^{-1}$, then $A = B^{-1}$ and inverse of an $n \times n$ matrix, if exists, is unique

Remark : $\rho(A) = n \iff \det(A) \neq 0$.

Ex - $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ Rank = 2 $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ linear independent $\therefore$ Rank = 2

$A = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ Rank = 3

Linear indep. Linear ind.

Q. Find $\rho(A)$ for the following matrices:

$\rightarrow A = \begin{pmatrix} 1 & 1 & 5 \\ 2 & 1 & 10 \\ 3 & 0 & 15 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 5 \end{pmatrix}$

$A = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

$\rightarrow$ Suppose A is a $n \times n$ matrix.

Define $N(A) := \{ \underset{\substack{\uparrow \\ \text{column vectors}}}{x} \in \mathbb{R}^n : Ax = \underset{0}{0} \} \subseteq \mathbb{R}^n$.

Thm: $N(A)$ is a linear subspace of $\mathbb{R}^n$.

Proof: Take $x, y \in N(A)$, $\alpha, \beta \in \mathbb{R}$

To show: $\alpha x + \beta y \in N(A)$

$x, y \in \mathbb{R}^n \Rightarrow \alpha x + \beta y \in \mathbb{R}^n$

Also, $Ax = Ay = 0$

$\therefore \underbrace{A(\alpha x + \beta y)}_{m \times n} = \underbrace{\alpha Ax + \beta Ay}_{n \times 1} = \alpha 0 + \beta 0 = 0$

$\Rightarrow \alpha x + \beta y \in N(A)$.

Defn: $N(A)$ is called the null space of A / Kernel of A .

$\nu(A) := \dim(N(A))$ is called the nullity of A .

Clearly $\nu(A) \leq n$.

Q. Given an $m \times n$ matrix A , how to compute $\nu(A)$.

Ex- Suppose $A_{2 \times 3} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

$$N(A) = \{ \tilde{x} \in \mathbb{R}^3 : A\tilde{x} = \underline{0} \}$$

$$= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

$$= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{pmatrix} x_1 + x_2 \\ x_2 + x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

$$= \left\{ (x_1, x_2, x_3)^T \in \mathbb{R}^3 : x_1 + x_2 = 0, x_2 + x_3 = 0 \right\}$$

It is easy to guess that $\nu(A) = \dim(N(A)) = 3 - 2 = 1$

Ex- $A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \end{bmatrix}$

$$N(A) = \{ \tilde{x} \in \mathbb{R}^3 : A\tilde{x} = \underline{0} \}$$

$$= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

$$= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{pmatrix} x_1 + x_2 \\ 2x_1 + 2x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

$$= \{ (x_1, x_2, x_3)^T \in \mathbb{R}^3 : x_1 + x_2 = 0 \}$$

$$\nu(A) = \dim(N(A)) = 3 - 1 = 2$$

In general

$$A\tilde{x} = \underline{0}$$

$\Rightarrow$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$m \times n$ $n \times 1$ $m \times 1$

Therefore $N(A) = \{ \tilde{x} \in \mathbb{R}^n : A\tilde{x} = \underline{0} \}$

$$= \left\{ (x_1, \dots, x_n)^T \in \mathbb{R}^n : \forall i = 1, 2, \dots, m, \sum_{j=1}^n a_{ij} x_j = 0 \right\}$$

Thm: Rank - Nullity thm.

for any $m \times n$ matrix A , $\rho(A) + \nu(A) = n$

$$\nu(A) = n - \rho(A)$$

Thm: $A x = b$
 $n \times n \quad n \times 1 \quad n \times 1$

If $m=n$, and A is non-singular, then
 $x = A^{-1} b$ is the unique soln. of the system (7)

Compute A^{-1} : If A is non-singular, then
 $A^{-1} = ((-1)^{i+j} M_{ji})_{1 \leq i, j \leq n} \cdot \frac{1}{|A|}$

where $M_{ji} = (j, i)^{th}$ minor of A := Determinant of $(n-1) \times (n-1)$ matrix obtained by deleting the j^{th} row and the i^{th} column.

$$|A| = a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}$$

Properties of determinants

- Exchange two rows $\rightarrow$ det changes sign
 - Multiply any rows by a constant $c \in \mathbb{R} \rightarrow$ det get multiplied by c
 - Add a scalar multiple of one row to another row $\rightarrow$ det remains unchanged
- $\rightarrow$ All these operations can be performed on columns also

Thm: For 2 $n \times n$ matrices A, B , we have
 $|AB| = |A||B|$

Thm: For any $n \times n$ matrix A , TFAE:
 $p(A) = n$ $|A| \neq 0$
 A is non-singular

Trace: For an $n \times n$ matrix A , $\text{tr}(A) = \text{trace}(A) := \sum_{i=1}^n a_{ii}$

For two $n \times n$ matrices A and B

$$\text{tr}(AB) = \text{tr}(BA)$$

$$\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$$

Defn: A $n \times n$ matrix A is called an idempotent matrix if $A^2 = A$. Ex - $A = I_n$

Thm: For an idempotent matrix A , $p(A) = \text{tr}(A)$

i. Suppose X is a $n \times p$ matrix. s.t. $X^T X$ is non-singular.

ii. Suppose J_n is $n \times n$ matrix with all entries 1. Using i, show that $E_n = \frac{1}{n} J_n$ is an idempotent matrix.

iii. Using ii, show that $p(E_n - \frac{1}{n} J_n) = n-1$

Partition Matrices

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{pmatrix}$$

$A_{21} \quad A_{22}$

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \text{ is called a partition matrix}$$

In general, let A be a $m \times n$ matrix and B be a $n \times p$ matrix. Take $1 \leq k \leq m-1$, $1 \leq l \leq n-1$, $1 \leq r \leq p-1$

we partition A as $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$

$\begin{matrix} m \times n \\ \begin{matrix} k \times l & k \times (n-l) \\ (m-k) \times l & (m-k) \times (n-l) \end{matrix} \end{matrix}$

$B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$

$\begin{matrix} n \times p \\ \begin{matrix} l \times r & l \times (p-r) \\ (n-l) \times r & (n-l) \times (p-r) \end{matrix} \end{matrix}$

Thm $AB_{(m \times p)} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}$

Remark: One can state this thm. for any partitioned matrix, not just with 2×2 blocks.

$$P_{4 \times 4} = \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} \quad Q_{4 \times 4} = \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{pmatrix}$$

$$P + Q = \begin{pmatrix} P_{11} + Q_{11} & P_{12} + Q_{12} \\ P_{21} + Q_{21} & P_{22} + Q_{22} \end{pmatrix}$$

1. a) Using Taylor's theorem, we get $\forall a, x \in \mathbb{R}$
 $k=1$

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(\eta)}{2!}(x-a)^2 \quad \text{for } \eta \in (a, x)$$

for $a=0$, we have $\forall x \in \mathbb{R}$

$$f(x) = f(0) + f'(0)x + \frac{f''(\eta)}{2}x^2$$

$$= \frac{f''(\eta)}{2}x^2 \quad \text{for some } \eta \text{ b/w } 0 \text{ and } x$$

$\therefore \forall x \in [-1, 1]$, we have

$$|f(x)| = \left| \frac{f''(\eta)}{2}x^2 \right| \leq \frac{1}{2}x^2 \leq \frac{1}{2}$$

f cont. on $[-1, 1] \Rightarrow f \in R[-1, 1]$.

Also, $\left| \int_{-1}^1 f(x) dx \right| \leq \int_{-1}^1 |f(x)| dx \leq \int_{-1}^1 \frac{1}{2} dx = 1$

$$\Rightarrow \left| \int_{-1}^1 f(x) dx \right| \leq 1$$

2. a) $\forall s \in (0, \infty)$ define $\phi_s: (0, \infty) \rightarrow (0, \infty)$ by
 $\phi_s(x) = x^{-s}, x \in (0, \infty)$

• Suppose $0 < s < 1, \phi_s \in R[\varepsilon, 1] \quad \forall \varepsilon \in (0, 1)$

Also, $\int_{\varepsilon}^1 \phi_s(x) dx = \int_{\varepsilon}^1 x^{-s} dx = \left[\frac{x^{1-s}}{1-s} \right]_{\varepsilon}^1 = \frac{1}{1-s} - \frac{\varepsilon^{1-s}}{1-s}$

$$s < 1 \Rightarrow 1-s > 0 \Rightarrow \varepsilon^{1-s} \rightarrow 0 \text{ as } \varepsilon \rightarrow 0^+$$

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^1 \phi_s(x) dx = \frac{1}{1-s} \Rightarrow \phi_s \in R[0, 1]$$

In fact $\int_{[0,1]} \phi_s(x) dx = \int_0^1 \phi_s(x) dx = \frac{1}{1-s}$

Suppose $s > 1$: clearly $\phi_s \in R[1, A] \quad \forall A > 1$

Also, $\int_1^A \phi_s(x) dx = \int_1^A x^{-s} dx = \left[\frac{x^{1-s}}{1-s} \right]_1^A = \frac{1-A^{1-s}}{s-1}$

$$s > 1 \Rightarrow s-1 > 0 \Rightarrow A^{1-s} \rightarrow 0 \text{ as } A \rightarrow \infty$$

$$\lim_{A \rightarrow \infty} \int_1^A \phi_s(x) dx = \frac{1}{s-1} \Rightarrow \phi_s \in R[1, \infty)$$

plane paper in
Page:
Date: / /

$g: (0, \infty) \rightarrow (0, \infty)$ defined by $g(x) = \frac{1}{x^3 + \sqrt[3]{x}}$ $x \in (0, \infty)$
is Riemann integrable on $(0, \infty)$.

$$\forall x \in (0, 1] \quad , \quad 0 < g(x) = \frac{1}{x^3 + \sqrt[3]{x}} \leq \frac{1}{\sqrt[3]{x}} \left[\begin{array}{l} x^3 + \sqrt[3]{x} \\ \geq \sqrt[3]{x} \end{array} \right]$$

Since, $0 < \frac{1}{3} < 1$, using 2(a) we get

$$\phi_{1/3}(x) \in R[0, 1]$$

$$\text{and } 0 < g(x) \leq \phi_{1/3}(x) \quad \forall x \in (0, 1]$$

therefore by comparison test, $g \in R(0, 1]$.

$$\forall x \in [1, \infty) \quad , \quad 0 \leq g(x) = \frac{1}{x^3 + \sqrt[3]{x}} \leq \frac{1}{x^3} \left[x^3 + \sqrt[3]{x} \geq x^3 \right]$$

$$\therefore 1 \leq 3 < \infty, \text{ using 2(a) we get } \phi_3(x)$$

$$\phi_3(x) \in R[1, \infty)$$

$$\text{and } 0 < g(x) \leq \phi_3(x) \quad \forall x \in [1, \infty)$$

therefore by comparison test, $g \in R[1, \infty)$

$$g \in R(0, 1] \cap R[1, \infty) \Rightarrow g \in R(0, \infty).$$

Try using Ratio Test.

$$5. \quad T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$$

$$T((x_1, x_2, x_3, x_4)^T) = (x_1 - 2x_2, x_2 + x_4, x_1 + 2x_4)^T$$

$$\Rightarrow \text{Take } \underline{u} = (u_1, u_2, u_3, u_4)^T \in \mathbb{R}^4,$$

$$\underline{v} = (v_1, v_2, v_3, v_4)^T \in \mathbb{R}^4, \text{ and } \alpha, \beta \in \mathbb{R}$$

$$\text{To show: } T(\alpha \underline{u} + \beta \underline{v}) = \alpha T(\underline{u}) + \beta T(\underline{v})$$

$$T(\alpha \underline{u} + \beta \underline{v}) = T(\alpha(u_1, u_2, u_3, u_4)^T + \beta(v_1, v_2, v_3, v_4)^T)$$

$$= T\left(\begin{pmatrix} \alpha u_1 + \beta v_1 \\ \alpha u_2 + \beta v_2 \\ \alpha u_3 + \beta v_3 \\ \alpha u_4 + \beta v_4 \end{pmatrix}\right)$$

$$= \begin{pmatrix} \alpha u_1 - 2\alpha u_2 \\ \alpha u_2 + \alpha u_4 \\ \alpha u_1 + 2\alpha u_4 \end{pmatrix} + \begin{pmatrix} \beta v_1 - 2\beta v_2 \\ \beta v_2 + \beta v_4 \\ \beta v_1 + 2\beta v_4 \end{pmatrix}$$

$$= T(\alpha(u_1, u_2, u_3, u_4)^T) + T(\beta(v_1, v_2, v_3, v_4)^T)$$

2. $\alpha T(\underline{u}) + \beta T(\underline{v})$
 $\therefore T$ is a linear map.

3-b. $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ is a linear map

Hence, $n=4$ and $m=3$ and hence A is a 3×4 matrix.

$$A_{*1} = 1^{\text{st}} \text{ column of } A = T \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$A_{*2} = 2^{\text{nd}} \text{ column of } A = T \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

$$A_{*3} = 3^{\text{rd}} \text{ column of } A = T \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$A_{*4} = 4^{\text{th}} \text{ column} = T \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 2 \end{pmatrix}$$

Claim A_{*1} and A_{*2} are linearly independent.

$$\alpha, \beta \in \mathbb{R} \text{ s.t. } \alpha A_{*1} + \beta A_{*2} = \underline{0}$$

$$\Rightarrow \alpha - 2\beta = 0, \quad \beta = 0, \quad \alpha = 0 \Rightarrow \alpha = \beta = 0$$

$\therefore A_{*1}$ and A_{*2} are linearly independent.

$\therefore A_{*3}$ is $\underline{0}$, so it won't increase the rank.

$$\text{Also, } \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \Rightarrow A_{*4} \text{ is a linear comb. of } A_{*1} \text{ and } A_{*2}$$

$$\therefore \rho(A) = 2.$$

4. $V := \mathbb{R}^4$, $S_1 := \{(v_1, v_2, v_3, v_4)^T \in \mathbb{R}^4 : v_1 - v_2 = v_3 - v_4 = 0\}$
 $S_2 := \{(u_1, u_2, u_3, u_4)^T \in \mathbb{R}^4 : u_1 + u_2 = u_3 + u_4 = 0\}$

to show $S_1 \subseteq \mathbb{R}^4 = V$
 (subsp)

Take $\underline{x} = (x_1, x_2, x_3, x_4)^T$, $\underline{y} = (y_1, y_2, y_3, y_4)^T \in S_1$

$$\Rightarrow x_1 - x_2 = x_3 - x_4 = 0 \quad \text{and} \quad y_1 - y_2 = y_3 - y_4 = 0$$

$$\underline{x} + \underline{y} = (x_1 + y_1, x_2 + y_2, x_3 + y_3, x_4 + y_4)^T$$

Take $\alpha, \beta \in \mathbb{R}$.

To show, $\alpha \underline{x} + \beta \underline{y} \in S_1$,

$$\alpha \underline{x} + \beta \underline{y} = \alpha \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \beta \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} \alpha x_1 + \beta y_1 \\ \alpha x_2 + \beta y_2 \\ \alpha x_3 + \beta y_3 \\ \alpha x_4 + \beta y_4 \end{pmatrix} \in \mathbb{R}^4$$

$$\therefore x_1 = x_2 \text{ and } y_1 = y_2$$

$$\Rightarrow \alpha x_1 = \alpha x_2 \text{ and } \beta y_1 = \beta y_2$$

$$\Rightarrow \alpha x_1 + \beta y_1 = \alpha x_2 + \beta y_2$$

$$\text{Similarly } \alpha x_3 + \beta y_3 = \alpha x_4 + \beta y_4$$

$$\therefore \begin{pmatrix} \alpha x_1 + \beta y_1 \\ \alpha x_2 + \beta y_2 \\ \alpha x_3 + \beta y_3 \\ \alpha x_4 + \beta y_4 \end{pmatrix} \in S_1 \Rightarrow S_1 \text{ is a subspace of } \mathbb{R}^4$$

Suppose V is a vector space and $\emptyset \neq S \subseteq V$. Then S is a linear subspace of V iff $\underline{x} + \underline{y} \in S \Rightarrow \underline{x} + \underline{y} \in S$,
 $\forall \alpha \in \mathbb{R}, \underline{x} \in S \Rightarrow \alpha \underline{x} \in S$.